

No Regret Policies for Online Control of Intelligent Reflecting Surfaces

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Abstract—Intelligent Reflecting Surfaces (IRSs) have been emerging as an extremely promising technology for smart control of radio environments. A crucial component behind the power of IRS is controlling the activation as well as phase angles of its meta-surfaces.

I. INTRODUCTION

Time frame of an IRS based communication system, can be typically divided into 3 segments.

- 1) At first it goes for pilot aided concatenated channel estimation (g_k, h_k) for each IRS element.
- 2) Next, optimal phase coefficient (ϕ_k and β_k) estimation is done using g_k , and h_k for each IRS element.
- 3) Finally using ϕ_k , input data estimation ($\hat{x}$) is done.

However, if the IRS elements are placed in a very close proximity then the channel parameters will become almost similar. So, switching on all the IRS elements at the same time instant will be an unnecessary wastage of power and resources. Because of this our motto is to switch on only a handful number of IRS elements. It leads to the obvious question that how to select those IRS elements. One possible solution is that the IRS elements showing 'strong' channel elements are to be chosen here.

Now, in order to search for the 'strong' IRS element, the best layman approach is to first switch on all IRS elements one by one, calculate the channel parameters individually, and finally choose the best ones. Although promising, but this approach will take a lot of time, and in a practical scenario where the user is moving fast, we do not have the liberty to invest a lot of time for IRS selection.

Therefore we propose to predict which IRS elements are to be switch on ($\beta_k = 1$), and then to estimate the corresponding channel properties only. There are two possible solutions to achieve this:

- 1) Channel estimation by individually turning on each IRS element. Next by solving an Optimization problem, we can select the 'strong' IRS element for estimating $\hat{x}$. Here the main issue is that the prediction of $\hat{x}$ is done, only after solving the optimization problem for active IRS element selection. As this is a serial approach, thus again it will take time, only worthy for slow fading.
- 2) As discussed earlier, serial approach is useless for fast fading channel. In order to mitigate this issue, we here proposed a parallel approach by introducing Online Caching Policy (henceforth is abbreviated as *OCP*).

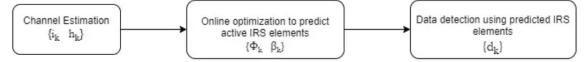


Fig. 1. Basic Communication System of IRS

The idea is described as follows:

For any time instant t

- a) Do the channel estimation (g_k, h_k) as previous.
- b) Now the *OCP* will do the optimization and after selection of active IRS element the corresponding information is fed to the next time instant ($t + 1$).
- c) Estimate $\hat{x}$ based on the phase element (ϕ_k) of the (strong) active ($\beta_k = 1$) IRS elements which are chosen by the *OCP* from the previous time instant ($t - 1$).

Now, here as the *OCP* is parallelly doing significant segment of the IRS estimation, thus this system is much faster, and we can implement it for fast fading channel

II. SYSTEM MODEL

Let the transmitted signal be x . Received signal in IRS is y , and final received signal in the receiver is r . Total number of IRS elements are N , and we are observing up-to T time instants. The reflection coefficient and the corresponding phase from the IRS are denoted by β , and ϕ .

Other notations are given below:

$t = 1, 2, \dots, T$

$k = 1, 2, \dots, N$

here

h_k = Input Channel gain for k 'th IRS element

g_k = Output Channel gain for k 'th IRS element

β_k = Reflection Coefficient of k 'th IRS element

ϕ_k = Phase of k 'th IRS element

So we can say that :

Incident wave in IRS is: $y_k = x h_k$

Reflected wave from IRS is: $\beta_k y_k$

Final Received Signal r is:

$$r = \left[\sum_{k=1}^N (g_k h_k \beta_k x e^{j\phi_k}) + n \right] \quad (1)$$

let $h_e = \beta_k g_k h_k e^{j\phi_k}$

Here, SNR can be written as:

$$\text{SNR} = \frac{|h_e|^2 P}{\sigma^2}$$

Thus SNR maximization can be done by increasing the value of $|h_e|^2$.

$$\begin{aligned} |h_e|^2 &= |\beta_k g_k h_k e^{j\phi_k}|^2 \\ &= |\beta_k d_k e^{j\phi_k}|^2 \end{aligned} \quad (2)$$

here $|d_k| = |g_k h_k|$

For simplicity purpose if we assume that reflection coefficient $(\beta_k) = 1$, then $|h_e|^2 = |d_k e^{j\phi_k}|^2$

Thus equation 2 can be maximized iff:

$$\left[\sum_{k=1}^N |d_k e^{j\phi_k}| \right] < \left[\sum_{k=1}^N |d_k| \right] \quad (3)$$

Now $d_k e^{j\phi_k} = |d_k| e^{j\phi_k}$

Thus from Eq.(3) we can write down:

$$e^{j\phi_k} e^{j\phi_k} = e^{mod(2\pi)} \quad (4)$$

which leads to:

$$\begin{aligned} \angle d_k + \angle \phi_k &= mod(2\pi) \\ \angle \phi_k &= -\angle d_k mod(2\pi) \end{aligned}$$

Finally ϕ_k can be expressed in terms of g_k and h_k as:

$$\angle \phi_k = -\angle g_k - \angle h_k mod(2\pi) \quad (5)$$

Equation (5) is the estimation of g_k and h_k , from ϕ_k , and vice versa.

When the channel is changing very slowly then we first estimate g_k and h_k (stage 1), next from them we estimate ϕ_k (stage 2), and finally $\hat{x}$ (estimated value) (stage 3).

Now, if the user is moving very fast, then the channel is changing very rapidly. Thus we don't have sufficient time to go through all the 3 stages. In this scenario we merge stage 1, and 2 in a single one. Here we first predict ϕ_k , and based upon it's value g_k and h_k are estimated. So if the estimation of ϕ_k is not done correctly then ultimately the prediction of $\hat{x}$ will also be erroneous. For fine tuning the value of ϕ_k , we feedback our $\hat{x}$, to match it, with the actual one over various time instants.

III. PROPOSED ALGORITHM

SM: Please populate the proposed algorithm here.

Now, from the above discussion, we can say that the difference between x (actual / best estimated) and $\hat{x}$ (predicted at current time instant, for k 'th IRS) will give us the Regret value. Mathematically it can be written as:

$$\begin{aligned} R_T &= \max_{\phi} \left[\sum_{t=1}^T \left| \sum_{k=1}^N (\beta_k d_{k,t} e^{j\phi_k}) \right|^2 \right] \\ &- \left[\sum_{t=1}^T \left| \sum_{k=1}^N (\beta_{k,t} d_{k,t} e^{j\phi_{k,t}}) \right|^2 \right] \end{aligned} \quad (6)$$

Thus, in order to get optimum result we have to minimize R_T

Does it means that we have to run the first segment of Eq (4) $\max_{\phi} \left[\sum_{t=1}^T \left| \sum_{k=1}^N (d_k e^{j\phi_k}) \right|^2 \right]$ for all combinations of β_k , and predicted ϕ_k to find out which combination of and gives us the maximum value?.

Also while doing the programming, shall we at first consider random values for β_k and ϕ_k ?

Condition 1: Initially for simplicity purpose we shall consider $\beta_k = 1$. That leads eq. 4 to:

$$\begin{aligned} R_T &= \max_{\phi} \left[\sum_{t=1}^T \left| \sum_{k=1}^N (d_{k,t} e^{j\phi_k}) \right|^2 \right] \\ &- \left[\sum_{t=1}^T \left| \sum_{k=1}^N (d_{k,t} e^{j\phi_{k,t}}) \right|^2 \right] \end{aligned} \quad (7)$$

Condition 2: This is the generalized version where $\phi_{k,t}$ can take any value between $\{0,1\}$. Because of this Eq.4 can be written as:

$$\begin{aligned} R_T &= \max_{\beta} \left[\sum_{t=1}^T \mathbb{E} \left[\sum_{k=1}^N (d_{k,t} \beta_k e^{j\phi_k})^2 \right] \right] \\ &- \sum_{t=1}^T \mathbb{E} \left[\sum_{k=1}^N (d_{k,t} \beta_{k,t} e^{j\phi_{k,t}})^2 \right] \end{aligned}$$

Where:

$$\mathbb{E} \left[\sum_{k=1}^N (\beta_k d_{k,t} e^{j\phi_k})^2 \right] = \mathbb{E} \left[\sum_{k,l} (\beta_k^* \beta_l d_{k,t}^* d_{l,t} e^{-j\phi_k} e^{j\phi_l}) \right] \quad (8)$$

or

$$\mathbb{E} \left[\sum_{k=1}^N (\beta_k d_{k,t} e^{j\phi_k})^2 \right] = \sum_{k,j} (\beta_k^2 d_{k,t}^2 \mathbb{E} [e^{-j\phi_k} e^{j\phi_l}]) \quad (9)$$

Lemma 1:

As the phase is uniformly distributed over $[0,2\pi]$, thus :

$$\mathbb{E} [e^{-j\phi_k} e^{j\phi_j}] = \frac{1}{2\pi} \left[\int_0^{2\pi} e^{j(\phi_k - \phi_l)} d\phi \right] \quad (10)$$

(i) if $k=l$ then eq. (L1) can be written as:

$$\mathbb{E} [e^{-j\phi_k} e^{j\phi_l}] = \frac{1}{2\pi} \left[\int_0^{2\pi} (1) d\phi \right] = 1 \quad (11)$$

(ii) i) If $k \neq l$ then eq.(L1) can be written as:

$$\begin{aligned} \mathbb{E} [e^{-j\phi_k} e^{j\phi_l}] &= \frac{1}{2\pi} \left[\int_0^{2\pi} e^{j(\phi_m)} d\phi \right] \\ &= \frac{1}{2j\pi} [1 - 1] = 0 \end{aligned} \quad (12)$$

where $\phi_m = \phi_l - \phi_k$

Using Lemma 1, eq.(4) can be re-written as:

$$R_T = \max_{\beta} \left[\sum_{t=1}^T \left| \sum_{k=1}^N (\beta_k d_{k,t}) \right|^2 \right] - \left[\sum_{t=1}^T \left| \sum_{k=1}^N (\beta_{k,t} d_{k,t}) \right|^2 \right] \quad (13)$$

That means the prediction of ϕ_k is not at all relevant for efficient channel estimation.

[But if we ignore it then how can we calculate g_k, h_k (eqn.(3b)) ? Or here also g_k, h_k have been taken randomly, and then fine tuning is done to minimize R_T].

Eq.(4c) is further written as:

$$R_T = \max_{\beta} \left[\sum_{t=1}^T \langle \beta^2, b_{t,k} \rangle \right] - \left[\sum_{t=1}^T \langle \beta^2, b_{t,k} \rangle \right] \quad (14)$$

Here $\beta_i \in 0,1$ and $b_{t,k} = |d_{k,t}|^2$

A. Lower and Upper Limit Analysis of Regret

As β_k can take only binary values (for On-Off) condition, thus equation 13 can be re-written as

$$R_T = \max_{\beta} \left[\sum_{t=1}^T \beta_k \sum_{k=1}^N |d_{k,t}|^2 \right] - \left[\sum_{t=1}^T \sum_{k=1}^N \beta_{k,t} |d_{k,t}|^2 \right] \quad (15)$$

or

$$R_T = \max_{\underline{\beta} \in \Delta_{N,K}} \langle \underline{\beta}, \sum_{t=1}^T \underline{d}_t \rangle - \sum_{t=1}^T \langle \underline{\beta}_t, \underline{d}_t \rangle \quad (16)$$

where $\Delta_{N,K}$ is the K active IRS elements out of a total of N elements. Mathematically, it can be represented as:

$$\Delta_{N,K} = \left\{ \underline{x} \in \{0,1\}^N : \sum_{i=1}^N x_i = K \right\} \quad (17)$$

and

$$\underline{d}_t = [|d_{1,t}|^2, \dots, |d_{N,t}|^2]^T$$

SM: Remember that we were having problems analyzing the FTPL algorithm. We were thinking about using FTRL using the k -experts paper as a reference. Write accordingly.

Now to apply the FTPL algorithm, we have defined the potential function $\phi_n(\underline{d})$ as:

$$\phi_n(\underline{d}) = \mathbb{E}_{\underline{\gamma}} \left[\max_{\underline{\beta} \in \Delta_{N,K}} \langle \underline{\beta}, \underline{d} + \eta \cdot \underline{\gamma} \rangle \right] \quad (18)$$

Now, the gradient of the above function can be written as:

$$\begin{aligned} \nabla (\phi_n(\underline{d}))_i &= \lim_{h_i \rightarrow 0} \frac{\phi_n(d_1, \dots, d_{i-1}, d_i + h_i, d_{i+1}, \dots, d_N) - \phi_n(d_1, \dots, d_N)}{h_i} \\ &= \lim_{h_i \rightarrow 0} \mathbb{E}_{\underline{\gamma}} \left[\frac{\hat{\beta}_i^* - \beta_i^*}{h_i} \right] \end{aligned} \quad (19)$$

where

$$\underline{\beta}^* = \arg \max_{\underline{\beta} \in \Delta_{N,K}} \langle \underline{\beta}, \underline{d} + \eta \cdot \underline{\gamma} \rangle$$

and

$$\hat{\beta}^* = \arg \max_{\underline{\beta} \in \Delta_{N,K}} \langle \underline{\beta}, \underline{d} + h_i \cdot \underline{e}_i + \eta \cdot \underline{\gamma} \rangle$$

Here $\underline{e}_i$ is the one hot k -dimensional array. The above formulation leads to the following inequalities:

$$\langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle \geq \langle \underline{\beta}, \underline{d} + \eta \cdot \underline{\gamma} \rangle \quad \forall \underline{\beta} \in \Delta_{N,K} \quad (20)$$

and

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \geq \langle \underline{\beta}, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \quad \forall \underline{\beta} \in \Delta_{N,K} \quad (21)$$

A simple inspection thus further leads us to establish the inequalities between $\underline{\beta}^*$ and $\hat{\beta}^*$, listed as follows:

$$\langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle \geq \langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle \quad (22)$$

and

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \geq \langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \quad (23)$$

The R.H.S of Equation 22 can be simplified as $\langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \beta_i^*$

This leads to

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \geq \langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \beta_i^* \quad (24)$$

It ultimately gives a simplified expression of $\nabla (\phi_n(\underline{d})) = \mathbb{E}_{\underline{\gamma}} [\underline{\beta}^*]$ Now, by taking the help of equation 22 and further simplifying the L.H.S of equation 23 we can deduce that

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \hat{\beta}_i^* \geq \langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \beta_i^*$$

This inequality further leads to

$$h_i \cdot \hat{\beta}_i^* \geq h_i \cdot \beta_i^* \quad (25)$$

Now as $h_i > 0$ thus $\hat{\beta}_i^* \geq \beta_i^*$. Next, if we consider $\hat{\beta}_i^* \neq \beta_i^*$ then, $\exists \delta > 0$, such that

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle = \langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle + \delta$$

or

$$\langle \hat{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle \geq \langle \underline{\beta}^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \beta_i^* + \delta \quad (26)$$

Equation 24 yields a slight modification over equation 23 as

$$h_i \cdot \hat{\beta}_i^* \geq h_i \cdot \beta_i^* + \delta \quad (27)$$

this further yields to

$$\hat{\beta}_i^* \geq \beta_i^* + \frac{\delta}{h_i}$$

the only condition, satisfying the above inequality is $\hat{\beta}_i^* = 1$, which leads to $h_i \geq \delta$.

We can examine the problem by introducing two variables (controlling parameters) corresponding to h_i . These are from equations 22, and 23, where the inequality is adjusted as:

$$\langle \hat{\beta}_i^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle = \langle \beta_i^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle + \delta_1$$

$$\langle \beta_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle = \langle \hat{\beta}_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + \delta_2$$

For obvious reasons as $\delta_1, \delta_2 > 0$, we can readjust and simplify the above equations as:

$$\langle \hat{\beta}_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \hat{\beta}_i^* = \langle \beta_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle + h_i \cdot \beta_i^* + \delta_1 + \delta_2$$

or

$$h_i \cdot \hat{\beta}_i^* = h_i \cdot \beta_i^* + \delta_1 + \delta_2 \quad (28)$$

Now, as $\beta_i^*, \hat{\beta}_i^* \in \{0, 1\}$, so for $h_i > 0$ equation 26 leads us to

$$\hat{\beta}_i^* = \beta_i^* + \frac{\delta_1 + \delta_2}{h_i} \quad (29)$$

The only possibility to satisfy the above is $\hat{\beta}_i^* = 1, \beta_i^* = 0$ as $\arg\max \beta = 1$. Putting these values in Equation 26 we get

$$h_i = \delta_1 + \delta_2$$

Thus

$$\begin{aligned} \langle \hat{\beta}_i^*, \underline{d} + \eta \cdot \underline{\gamma} + h_i \cdot \underline{e}_i \rangle - \langle \beta_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle &= h_i + \langle \hat{\beta}_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle - \langle \beta_i^*, \underline{d} + \eta \cdot \underline{\gamma} \rangle \\ &= h_i - \delta_2 \end{aligned}$$

Finally putting this value into the corollary of Equation 18, we get

$$\begin{aligned} \nabla (\phi_n(\underline{d}))_i &= \lim_{h_i \rightarrow 0} \frac{h_i - \delta_2}{h_i} \\ &= \lim_{\delta_1, \delta_2 \rightarrow 0} \frac{\delta_1}{\delta_1 + \delta_2} = 0 \end{aligned}$$

Next, we have slightly modified the expression of β_i^* and pitched a similar kind of term $\underline{\beta}_t$, based on the best prediction up to the last time instant. Mathematically it can be expressed as

$$\underline{\beta}_t = \arg \max_{\beta \in \Delta_{N,K}} \langle \beta, \underline{D}_{t-1} + \eta \cdot \underline{\gamma} \rangle \quad (30)$$

Where $\underline{D}_t = \sum_{s=1}^t \underline{d}_s$, and as we are following *FTPL* algorithm, thus $\nabla (\phi(\underline{D}_{t-1})) = \mathbb{E} [\underline{\beta}_t]$. By application of the Taylor Series of second order, the potential function at t 'th time instant can be written as:

$$\begin{aligned} \phi(\underline{D}_t) &= \phi(\underline{D}_{t-1}) + (\underline{D}_t - \underline{D}_{t-1})^\top \nabla \phi(\underline{D}_{t-1}) \\ &\quad + \frac{1}{2} (\underline{D}_t - \underline{D}_{t-1})^\top \nabla^2 \phi(\tilde{\underline{D}}_t) (\underline{D}_t - \underline{D}_{t-1}) \end{aligned} \quad (31)$$

where $\tilde{\underline{D}}_t \in [\underline{D}_{t-1}, \underline{D}_t]$. Further simplification of equation 30 leads us to

$$\begin{aligned} \phi(\underline{D}_t) &= \phi(\underline{D}_{t-1}) + \underline{d}_t^\top \mathbb{E} [\underline{\beta}_t] \\ &\quad + \frac{1}{2} \underline{d}_t^\top \nabla^2 \phi(\tilde{\underline{D}}_t) \underline{d}_t \end{aligned}$$

Summing over the total time instant T , we get

$$\begin{aligned} \phi(\underline{D}_T) &= \phi(\underline{D}_1) + \sum_t \underline{d}_t^\top \mathbb{E} [\underline{\beta}_t] \\ &\quad + \sum_t \frac{1}{2} \underline{d}_t^\top \nabla^2 \phi(\tilde{\underline{D}}_t) \underline{d}_t \end{aligned} \quad (32)$$

The application of *Jensen's inequality* in equation 17 with the mean of 0 gives us

$$\phi(\underline{D}_T) \geq \max_{\underline{\beta}} \langle \underline{\beta}, \underline{D}_T \rangle \quad (33)$$

Finally, by using equations 31, and 32, we can deduce the upper bound of the Regreat function, formulated as:

$$\begin{aligned} \mathbb{E}[R_t] &= \max_{\underline{\beta}} \langle \underline{\beta}, \underline{D}_T \rangle - \mathbb{E} \left[\sum_t \underline{d}_t^\top \underline{\beta}_t \right] \\ &\leq \phi(\underline{D}_1) + \sum_t \frac{1}{2} \underline{d}_t^\top \nabla^2 \phi(\tilde{\underline{D}}_t) \underline{d}_t \end{aligned} \quad (34)$$

For K number of IRS elements, and $\text{var}[\gamma] = \sigma$ the upper bound of $\phi(\underline{D}_1)$ can be written as (using Massart's Lemma):

$$\phi(\underline{D}_1) \leq \eta \sigma \sqrt{2 \ln(K \cdot c)}$$

and

$$\begin{aligned} \left[\nabla^2 \phi(\tilde{\underline{D}}_t) \cdot \underline{d}_t \right]_{i,j} &= \frac{1}{\eta} \mathbb{E} \left[\hat{\beta}_i(\tilde{\underline{D}}_t + \eta \underline{\gamma}_i) \cdot \gamma_j \right] \\ &= \frac{1}{\eta} \mathbb{E} \left[\gamma_j \cdot \mathbf{1} \left\{ \hat{\beta}_i(\tilde{\underline{D}}_t + \eta \underline{\gamma}_i) = 1 \right\} \right] \end{aligned}$$

Although promising, this existing *FTPL* algorithm suffers from two major drawbacks:

- All the existing problems deal with *OneHotArray*, but in the case of IRS, most of the time, multiple elements will become active. Which forbids the existing methodologies to be effective.
- Second, and most significant drawback is to set the limit of $\underline{D}_t$. Any existing literature does not show any insight of how to estimate it.

Because of this, we have moved towards a SAGE-based FTRL algorithm to overcome the challenges. As given in [], this algorithm removes the first issue by considering multiple entries that are active at the same time. Now, before going deep, let us formulate the basis of the SAGE-based algorithm and how to formulate the upper bound.

So, in the beginning, from Equation 17 we can say that as we select only those N -dimensional columns that have exactly K entries equal to one. thus $|\Delta_{N,K}| = \binom{N}{K}$, and $\underline{\beta}_t \equiv S_t$. This ultimately leads to the prediction of S_t at each time such that

$$P(S_t = S) = p_t(S), \quad \text{where } S \text{ is a } K\text{-set.} \quad (35)$$

Without the loss of generality

$$\begin{aligned} \mathbb{E} \left[\sum_{t=1}^T \langle \underline{\beta}_t, \underline{d}_t \rangle \right] &= \sum_{t=1}^T \langle \mathbb{E} [\underline{\beta}_t], \underline{d}_t \rangle \\ &= \sum_{t=1}^T \left(\sum_{S \in \mathcal{f}_{N,K}} p(S) \langle \mathbf{1}_S, \underline{d}_t \rangle \right) \\ &= \sum_{t=1}^T \langle \underline{p}_t, \underline{d}_t \rangle \end{aligned} \quad (36)$$

where $\mathcal{I}_{N,K} = \{S \subset [N] : |S| = K\}$.

Using the above equality, we can calculate $\mathbb{E}[R_T]$ from Equation 16 as:

$$\mathbb{E}[R_T] = \max_{\underline{p} \in \Delta_{N,K}} \left(\sum_{t=1}^T \langle \underline{p}, \underline{d}_t \rangle \right) - \sum_{t=1}^T \langle \underline{p}_t, \underline{d}_t \rangle \quad (37)$$

As evident from the above equation, our main job is now to estimate $\underline{p}_t$ such that the Regret is minimal. This is obtained by implementing the FTRL algorithm. The key difference between FTPL and FTRL is that FTPL adds random noise to promote exploration, while FTRL adds a regularization step to control model complexity and promote sparseness. As mentioned in [], we have also used Entropy as the regularizer here. This gives us $\underline{p}_t$ as:

$$\underline{p}_t = \arg \max_{\underline{p} \in \Delta_{N,K}} \left[\left\langle \underline{p}, \sum_{s=1}^{t-1} \underline{d}_s \right\rangle - \frac{1}{\eta} \sum_{i=1}^N p_i \ln(p_i) \right] \quad (38)$$

Application of the above policy leads us to find the upper bound of the Regret as

$$\text{Regret}_T \leq \frac{K \ln(N/K)}{\eta} + 2\eta \sum_{t=1}^T \|\underline{d}_t\|_{K,\infty}^2 \quad (39)$$

Now our job is reduced to find the upper bound of $|d_{t,i}|$. If, say $|d_{t,i}| \leq B \quad \forall t, i$, then the upper bound will become $\frac{K \ln(N/K)}{\eta} + 2\eta KTB$. Upon maximizing it over η , the upper bound is finally reduced as:

$$\text{Regret}_T \leq 2K \sqrt{2TB \ln(N/K)}$$

Thus the complexity of Regret_T is $\mathcal{O}\left(k \sqrt{T \ln(N/K)}\right)$

Limitation of taking iid nature of channel responses: From the previous discussions, we know that $d_{t,i} = |h_{t,i}|^2 \cdot |g_{t,i}|^2$. Now for any particular IRS element i , those channel responses are independent, i.e. $g_{t,i} \perp h_{t,i}$, but for any coherent time instant t , this cannot be said about adjoint IRS elements. Thus, let us say for neighboring IRS elements i, j we can have conditions like

$$h_{t,i} \not\perp h_{t,j} \text{ and } g_{t,i} \not\perp g_{t,j}; i \neq j$$

However, for individual coherence time, the channel responses for the same IRS elements are independent

$$h_{t,i} \perp h_{s,i} \text{ and } g_{t,i} \perp g_{s,i}; t \neq s$$

B. Revised Analysis

Here, $\|d_{t,i}\|_{K,\infty}^2$ stands for the summation of the K largest components. Thus, mathematically we can write:

$$\|d_{t,i}\|_{K,\infty}^2 = \max_{|S|=K} \sum_{i \in S} \|g_{t,i}\|^2 \cdot \|h_{t,i}\|^2 \quad (40)$$

For notational convenience, we shall henceforth denote $\xi_t = \|d_{t,i}\|_{K,\infty}^2$. Our objective is to find a high-probability upper

bound a for the quantity of interest. Mathematically, this can be expressed as

$$\mathbb{P} \left(\sum_{t=1}^T \xi_t \geq a \right) \leq \delta.$$

Assuming that the channel parameters are independent and identically distributed (i.i.d.), the above expression can be rewritten as

$$\mathbb{P} \left(\sum_{t=1}^T (\xi_t - \mathbb{E}[\xi_t]) \geq a - T \cdot \mathbb{E}[\xi] \right) \leq \delta. \quad (41)$$

Next, by defining $X_t = \xi_t - \mathbb{E}[\xi_t]$ and $b = a - T \cdot \mathbb{E}[\xi]$, the above equation simplifies to

$$\mathbb{P} \left(\sum_{t=1}^T X_t \geq b \right) \leq \delta,$$

where $\mathbb{E}[X_t] = 0$.

Since g and h follow Gaussian distributions, the quantities $|g|^2$ and $|h|^2$ follow exponential distributions. Thus it makes the distribution of ξ becomes *sub-exponential* in nature.

As sub-exponential distributions possess *heavier tails* than the exponential distribution, traditional analyses based on the Martingale Exponential Inequality are not applicable in this case, as they rely on the existence of finite moment generating functions (MGFs), which do not hold here.

To address this limitation, we consider **Nagaev's inequality** (see [ref]). In particular, applying **Corollary 1.8** from [ref], with a moment of order $t = 2$, we obtain the following upper bound:

$$\mathbb{P} \left(\sum_{t=1}^T X_t \geq b \right) \leq 4A_2^+ b^{-2} + \exp \left(-\frac{e^{-2} b^2}{8B^2} \right), \quad (42)$$

Here, the constant A_2^+ is defined as

$$A_2^+ = \sum_{t=1}^T \int_{X \geq 0} X_t^2 dF_t(X),$$

where $F_t(X)$ denotes the cumulative distribution function (CDF) of X_t .

Similarly, B^2 is defined as

$$B^2 = \sum_{t=1}^T (\text{Var}[X_t])^2,$$

This leads to the final bound:

$$\delta = 4A_2^+ b^{-2} + \exp \left(-\frac{e^{-2} b^2}{8B^2} \right).$$

Calculation of b

The value of b can be estimated through two approaches:

First Approach (Large- b Approximation): If we assume that b is sufficiently large, the exponential term becomes negligible, i.e.,

$$\exp \left(-\frac{e^{-2} b^2}{2B^2} \right) \approx 0,$$

and the expression for δ simplifies to:

$$\delta \approx 4A_2^+ b^{-2} \Rightarrow b \approx 2\sqrt{\frac{A_2^+}{\delta}}. \quad (43)$$

We avoid considering very small values of b because, without loss of generality, we can assume that heavy-tailed distributions have their tails at values significantly distant from the mean. Mathematically, also for a very small value of b , the exponential term is approximately equal to 1, leading to:

$$\delta \approx 1 + 4A_2^+ b^{-2} \Rightarrow b \approx 2\sqrt{\frac{A_2^+}{\delta - 1}}.$$

This leads to an imaginary value of b , as $\delta \leq 1$, which is unfeasible.

Second Approach (Uniform Upper Bound): Alternatively, we can upper bound the exponential term by the polynomial term. Specifically, we consider

$$\exp\left(-\frac{e^{-2}b^2}{8B^2}\right) \leq 4A_2^+ b^{-2},$$

so that the entire expression for δ can be upper bounded as

$$\delta \leq 8A_2^+ b^{-2} \Rightarrow b = 2\sqrt{\frac{2A_2^+}{\delta}}. \quad (44)$$

As our target is to check for the maximum range of a heavy-tailed distribution, we prefer the value of b obtained from Equation (44) as it is larger than the one derived in Equation (43). Thus for all further calculations we consider b as:

$$b = 2\sqrt{\frac{2A_2^+}{\delta}}.$$

Lemma 1: Let $X_1, X_2, \dots, X_K$ be i.i.d. real-valued random variables, and let $\lambda \geq 1, \eta > 0$. Then we can write that:

$$\begin{aligned} \mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] &= \frac{1}{\eta} \left(\mathbb{E} \left[\left(\eta \cdot \max_{1 \leq i \leq K} X_i \right)^\lambda \right] \right)^{1/\lambda} \\ &\leq \left(\mathbb{E} \left[\max_{1 \leq i \leq K} X_i^\lambda \right] \right)^{1/\lambda} \\ &\leq \left(\sum_{i=1}^K \mathbb{E}[X_i^\lambda] \right)^{1/\lambda} \\ &= (K \cdot \|X\|_\lambda^\lambda)^{1/\lambda} \quad (\text{since all } X_i \text{ are i.i.d.}) \\ &= K^{1/\lambda} \cdot \|X\|_\lambda. \end{aligned}$$

The final inequality thus can be written as:

$$\mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] \leq K^{1/\lambda} \cdot \|X\|_\lambda \quad \text{for all } \lambda \geq 1. \quad (45)$$

Now, to fit this into our algorithm, let us take a closer look at our variable of interest. As described above, the random variable ξ possesses a heavy-tailed distribution. Consequently, both $\xi - \mathbb{E}[\xi]$ and each X_i (constructed from ξ) also exhibit heavy tails.

Due to this heavy-tailed behavior, we invoke a *Sub-Weibull* based analysis for X_i . In particular, if $X \sim \text{SubW}(\theta)$ for some $\theta \in (0, 1)$, then for all $\lambda \geq 1$, we can model $\|X\|_\lambda$ as [].

$$\|X\|_\lambda \leq C \cdot \lambda^\theta,$$

where $C > 0$ is a constant depending on the sub-Weibull norm of X .

Substituting it into equation (45) we get

$$\begin{aligned} \mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] &\leq K^{1/\lambda} \cdot \|X\|_\lambda \\ &\leq C \cdot K^{1/\lambda} \cdot \lambda^\theta \\ &= C \cdot \exp \left(\theta \ln \lambda + \frac{1}{\lambda} \ln K \right). \end{aligned}$$

Now, let us find an upper limit for the expectation term. A close inspection of the expression

$$\mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] \leq C \cdot \exp \left(\theta \ln \lambda + \frac{1}{\lambda} \ln K \right) \quad (46)$$

reveals that the only variable term is λ . Therefore, we define the function:

$$f(\lambda) = \theta \ln \lambda + \frac{1}{\lambda} \ln K,$$

and seek to minimize $f(\lambda)$ over $\lambda \geq 1$.

By applying gradient descent, we find the minimizer point as:

$$\lambda^* = \frac{\ln K}{\theta}.$$

Substituting this optimal λ^* into equation(46) gives the bound as:

$$\mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] \leq C \cdot \left(\frac{\ln K}{\theta} \right)^\theta \cdot e^\theta. \quad (47)$$

As for the sub-exponential case, we have $\theta = 1$. Substituting this into equation (47), the simplified upper bound is:

$$\mathbb{E} \left[\max_{1 \leq i \leq K} X_i \right] \leq C \cdot \ln K \cdot e.$$

Here, we assume that the sequences $\{g_t\}_{t=1}^T$ and $\{h_t\}_{t=1}^T$ are independent. However, since we are considering a fast-fading channel, these random variables are not identically distributed over time. Instead, their power profiles vary across time instants.

In the case of a Rayleigh fading channel, this results in different spreading-parameter (σ) profiles at different time instances.

Now, let us apply **Nagaev's inequality** along with **Lemma 1** to compute an upper bound on $\sum_{t=1}^T \xi_t$. In this regard, at first we define the quantity A as follows:

$$A = \sum_{t=1}^T \xi_t = \sum_{t=1}^T X_t + T' \cdot \mathbb{E}[\xi_1], \quad (48)$$

CALCULATION OF T'

Let us consider the random variables $|g|$ and $|h|$, both following Rayleigh distributions with parameter σ , i.e.,

$$|g| \sim \text{Rayleigh}(\sigma), \quad |h| \sim \text{Rayleigh}(\sigma).$$

Then, the squared magnitudes follow exponential distributions:

$$|g|^2 \sim \text{Exp}(2\sigma^2), \quad |h|^2 \sim \text{Exp}(2\sigma^2).$$

Now, define

$$\xi_1 = \sum_t |g_1|^2 \cdot |h_1|^2.$$

Taking the expectation and using the independence of $|g_1|$ and $|h_1|$, we get the following.

$$\mathbb{E}[\xi_1] = \sum_t \mathbb{E}[|g_1|^2] \cdot \mathbb{E}[|h_1|^2] = \sum_t (2\sigma_t^2)^2.$$

That gives the final expression:

$$T' = 4 \sum_t \sigma_t^4 \quad (49)$$

Now, let us solve Equation (42), where

$$A_2^+ = T'' \cdot \mathbb{E}[X_1^2 \cdot \mathbf{1}\{X_1 \geq 0\}], \quad \text{and} \quad B^2 = T'' \cdot \mathbb{E}[X_1^2].$$

Following similar steps as in the previous section, T'' can be computed as

$$T'' = 16 \sum_t \sigma_t^8 \quad (50)$$

We now consider the probability bound:

$$\mathbb{P}\left(\sum_t X_t \geq T''b\right) \leq \frac{4A_2^+}{(T''b)^2} + \exp\left(-\frac{e^{-2}(T''b)^2}{8B^2}\right).$$

We define the normalized constants:

$$a_1 = \frac{A_2^+}{T''}, \quad a_2 = \frac{B^2}{T''}.$$

Substituting into the bound, we obtain:

$$\mathbb{P}\left(\sum_t X_t \geq T''b\right) \leq \frac{4a_1}{b^2} + \exp\left(-\frac{e^{-2}b^2}{8a_2}\right). \quad (51)$$

Next, by utilizing the inequality $e^{-x} < \frac{1}{x}$, the right-hand side of Equation (51) can be further upper bounded as follows:

$$\begin{aligned} \frac{4a_1}{(T''b)^2} + \exp\left(-\frac{e^{-2}(T''b)^2}{8a_2}\right) &< \frac{4a_1}{(T''b)^2} + \frac{8a_2}{e^{-2}(T''b)^2} \\ &= \frac{1}{T''} \left(\frac{4a_1}{b^2} + \frac{8a_2}{e^{-2}b^2} \right). \end{aligned}$$

Let us define a constant:

$$C = \frac{4a_1}{b^2} + \frac{8a_2}{e^{-2}b^2},$$

Now, since $a_2 \geq a_1$, we can bound C as follows:

$$C \leq \frac{4a_2}{b^2} (1 + 2e^2).$$

Thus the entire bound becomes:

$$\frac{C}{T''} \leq \delta.$$

This implies:

$$T'' \geq \frac{C}{\delta}.$$

Substituting this into Equation (51), we obtain a simplified bound:

$$\mathbb{P}\left(\sum_t \xi_t \geq T''b + T' \cdot \mathbb{E}[\xi_1]\right) \leq \delta,$$

or equivalently,

$$\mathbb{P}\left(\sum_t \xi_t \leq T''b + T' \cdot \mathbb{E}[\xi_1]\right) \geq 1 - \delta. \quad (52)$$

where

$$T' \geq \sqrt{\frac{C}{\delta}}.$$

For simplicity, let us define

$$a_4 = \mathbb{E}[\xi_1], \quad \text{and} \quad a_3 = T''b + T'a_4.$$

Since b is a user-defined parameter, the evaluation of Equation (52) totally depends on the computation of a_2 and a_4 .

In the following sections, we will proceed with the detailed calculation of these two parameters.

Calculation of a_4

We defined

$$a_4 = \mathbb{E}[\xi_1] = \mathbb{E}\left[\max_{\substack{S \subseteq \{1, \dots, T\} \\ |S|=K}} \sum_{i \in S} |h_i|^2 |g_i|^2\right].$$

Let us denote:

$$Y_S = \sum_{i \in S} |h_i|^2 |g_i|^2.$$

Thus, we can rewrite:

$$\mathbb{E}[\xi_1] = \mathbb{E}\left[\max_{\substack{S \subseteq \{1, \dots, T\} \\ |S|=K}} Y_S\right].$$

Now, by applying Equation (45) from Lemma 1, we obtain the following upper bound:

$$\mathbb{E}[\xi_1] \leq K^{1/\lambda} \cdot (\mathbb{E}[Y_S^\lambda])^{1/\lambda}, \quad \text{for } \lambda > 0. \quad (53)$$

Continuing from the previous step, we use the property of the ℓ_λ -norm to further bound:

$$\mathbb{E}[\xi_1] \leq K^{1/\lambda} \left\| \sum_{i \in S} |h_i|^2 |g_i|^2 \right\|_\lambda.$$

By applying the triangle inequality, we get:

$$\left\| \sum_{i \in S} |h_i|^2 |g_i|^2 \right\|_\lambda \leq \sum_{i \in S} \| |h_i|^2 |g_i|^2 \|_\lambda.$$

Since $|S| = K$, and assuming identical distributions across i , (for any particular time instant, the IRS elements are i.i.d. in nature) we have:

$$\left\| \sum_{i \in S} |h_i|^2 |g_i|^2 \right\|_\lambda \leq K \| |h|^2 |g|^2 \|_\lambda.$$

Thus,

$$\mathbb{E}[\xi_1] \leq K^{1/\lambda} \cdot K \cdot \| |h|^2 |g|^2 \|_\lambda.$$

Now, using the definition of the ℓ_λ -norm:

$$\| |h|^2 |g|^2 \|_\lambda = \left(\mathbb{E} \left[(|h|^2 |g|^2)^\lambda \right] \right)^{1/\lambda} = (\mathbb{E}[|h|^{2\lambda}] \cdot \mathbb{E}[|g|^{2\lambda}])^{1/\lambda}.$$

Combining everything, we obtain:

$$\mathbb{E}[\xi_1] \leq K^{(1+\lambda)/\lambda} (\mathbb{E}[|h|^{2\lambda}] \cdot \mathbb{E}[|g|^{2\lambda}])^{1/\lambda}.$$

Now, since both $|g|$ and $|h|$ are Rayleigh faded with the same parameter σ , their squared magnitudes $|g|^2$ and $|h|^2$ are independent and follow the exponential distribution with parameter $2\sigma^2$.

Since $|g| \stackrel{d}{=} |h|$, the above equation becomes:

$$\mathbb{E}[\xi_1] \leq K^{(1+\lambda)/\lambda} (\mathbb{E}[|h|^{2\lambda}])^{2/\lambda}. \quad (54)$$

Now, as $|h| \sim \text{Rayleigh}(\sigma)$, thus we can easily calculate:

$$\mathbb{E}[|h|^{2\lambda}] = \sigma^{2\lambda} \cdot 2^\lambda \cdot \Gamma(\lambda + 1).$$

Using the upper bound from Stirling's approximation:

$$\Gamma(\lambda + 1) \leq \lambda^\lambda e^{-\lambda} \sqrt{2\pi\lambda},$$

we obtain:

$$\mathbb{E}[|h|^{2\lambda}] \leq \sigma^{2\lambda} \cdot 2^\lambda \cdot \lambda^\lambda e^{-\lambda} \sqrt{2\pi\lambda}.$$

Thus, we can finally deduce the upper bound on a_4 as:

$$a_4 = \mathbb{E}[\xi_1] \leq C_1 \cdot K^{\frac{1+\lambda}{\lambda}} \cdot \lambda^2, \quad (55)$$

where the constant C_1 is given by:

$$C_1 = 4\sigma^4 \cdot e^{-2} \cdot e^{1/e}.$$

We can further proceed to search for the upper bound of a_4 by optimizing the parameter λ . To facilitate this, we define the function $f(\lambda)$ as follows:

$$f(\lambda) = \ln(a_4) = \ln(C_1) + \left(\frac{1+\lambda}{\lambda} \right) \ln K + 2 \ln(\lambda).$$

The critical point (i.e., the minimizer) λ^* of the function $f(\lambda)$ can be easily determined as

$$\lambda^* = \frac{\ln K}{2}.$$

Substituting this value into equation (55), we obtain a tight upper bound for a_4 as:

$$a_4 = \mathbb{E}[\xi_1] \leq \frac{C_1}{4} \cdot K^{\frac{1+\ln K}{\ln K}} \cdot (\ln K)^2. \quad (56)$$

Calculation of a_2 :

Recall that

$$a_2 = \mathbb{E}[X_1^2] = \mathbb{E}[(\xi_1 - \mathbb{E}[\xi_1])^2] = \text{Var}(\xi_1).$$

Let us introduce another term: $u_i = |h_i|^2 |g_i|^2$. Let m and v denote the mean and variance of u , respectively.

For simplicity, we define

$$z = \xi_1 = \max_{|S|=K} \sum_{i \in S} u_i. \quad (57)$$

Now, we apply the Efron–Stein inequality to calculate the upper bound on the variance of z . Mathematically, the Efron–Stein inequality states that

$$\text{Var}(z) \leq \frac{1}{2} \sum_{i=1}^S \mathbb{E}[(z - z'_i)^2], \quad (58)$$

where $z = f(u_1, u_2, \dots, u_S)$, and $z'_i = f(u_1, \dots, u_{i-1}, u'_i, u_{i+1}, \dots, u_S)$, with u'_i being an independent copy of u_i drawn from the same distribution.

Let S^* denote the subset of indices that maximizes the sum in the definition of z . That is, from Equation (57), we can express z and z'_j as follows:

$$z = \sum_{i \in S^*} u_i = u_j + \sum_{\substack{i \in S^* \\ i \neq j}} u_i.$$

By replacing u_j with its independent copy u'_j , we obtain

$$z'_j = \sum_{i \in S^*} u_i = u'_j + \sum_{\substack{i \in S^* \\ i \neq j}} u_i.$$

Now, by subtracting z'_j from z , and considering the case when $j \in S^*$ (i.e., the maximum is achieved when the change occurs only in the j -th coordinate), we observe that

$$|z - z'_j| \leq |u_j - u'_j|.$$

Therefore, from Equation (58), we can write

$$\text{Var}(z) \leq \frac{1}{2} \sum_{j=1}^S \mathbb{E}[(z - z'_j)^2] \leq \frac{1}{2} \sum_{j=1}^S \mathbb{E}[(u_j - u'_j)^2]. \quad (59)$$

Since u_j and u'_j are drawn independently from the same distribution, they have the same mean m and variance v . Substituting this we have:

$$\mathbb{E}[(u_j - u'_j)^2] = 2v.$$

Therefore, the variance of z (i.e., ξ_1) is:

$$\text{Var}(z) = \text{Var}(\xi_1) \leq \frac{1}{2} \sum_{i \in S} 2v = Kv. \quad (60)$$

Calculation of v

Let $G = |g|^2$ and $H = |h|^2$. Then the product $u = G \cdot H$. Since G and H are independent random variables, we have:

$$\mathbb{E}[u] = \mathbb{E}[G \cdot H] = \mathbb{E}[G] \cdot \mathbb{E}[H], \quad \mathbb{E}[u^2] = \mathbb{E}[G^2 \cdot H^2] = \mathbb{E}[G^2] \cdot \mathbb{E}[H^2].$$

As g and h follow a Rayleigh distribution with parameter σ , it is easy to find out that:

$$\mathbb{E}[G] = \mathbb{E}[H] = 2\sigma^2, \quad \text{Var}(G) = \text{Var}(H) = 4\sigma^4.$$

So, the variance of $u = G \cdot H$ is computed as:

$$\text{Var}(u) = \mathbb{E}[G^2] \cdot \mathbb{E}[H^2] - (\mathbb{E}[G] \cdot \mathbb{E}[H])^2. \quad (61)$$

We already know:

$$\mathbb{E}[G] = \mathbb{E}[H] = 2\sigma^2, \quad \text{Var}(G) = \text{Var}(H) = 4\sigma^4.$$

So,

$$\mathbb{E}[G^2] = \text{Var}(G) + (\mathbb{E}[G])^2 = 4\sigma^4 + (2\sigma^2)^2 = 8\sigma^4,$$

and similarly,

$$\mathbb{E}[H^2] = 8\sigma^4.$$

Substituting it Equation(61), we get:

$$v = \text{Var}(u) = 8\sigma^4 \cdot 8\sigma^4 - (2\sigma^2 \cdot 2\sigma^2)^2 = 64\sigma^8 - 16\sigma^8 = 48\sigma^8.$$

Finally, by substituting the value of $v = \text{Var}(u) = 48\sigma^8$ into Equation (60), we obtain the upper bound for a_2 as:

$$a_2 = \text{Var}(\xi_1) \leq 48K\sigma^8. \quad (62)$$

At different time instants, the system may select different subsets of IRS elements to compute

$$\max_{\substack{S \subseteq \{1, \dots, N\} \\ |S|=K}} \sum_{i \in S} u_i.$$

There are two main approaches to handle this situation:

1) **Higher-Order Effron–Stein Inequality:**

In this approach,

$$\text{Var}(z) = \sum_S \text{Inf}_S(f),$$

where $\text{Inf}_S(f)$ denotes the influence of the set S on the function f .

Although promising, this method requires knowledge of higher-order interactions, which are generally intractable and computationally expensive.

2) **Single-Element Change Effron–Stein Inequality:**

Let

$$\xi_t = \max_{\substack{S \subseteq \{1, \dots, N\} \\ |S|=K}} \sum_{i \in S} u_{t,i}, \quad z = \sum_{t=1}^T \xi_t.$$

In the worst-case scenario, all elements may be replaced. Define

$$\Delta_t = f(u_{t,1}, \dots, u_{t,N}) - f(u'_{t,1}, \dots, u'_{t,N}),$$

then the variance can be bounded as

$$\text{Var}(\xi_t) \leq \frac{1}{2} \mathbb{E}[\Delta_t^2].$$

However, evaluating Δ_t is generally quite challenging. However, by assuming that each $u_{t,i}$ has variance v , and by applying linearity and sub-additivity of expectation, we can approximate

$$\mathbb{E}[\Delta_t^2] K \cdot \mathbb{E}[(u_{t,i} - u'_{t,i})^2] = 2Kv = K\text{Var}(u),$$

which matches the previously derived result.

IV. NUMERICAL SIMULATIONS

V. CONCLUSION

REFERENCES